Smart Cat Litter *Box Monitor*

Passive health tracking for cats through visit, weight, and pattern monitoring.

TEAM

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TEAM

Team SmartLitterBox

GITHUB

https://github.com/MengqiS21/TECHIN515_final_project_smart_litter_box

The Problem & Who It Affects

THE PROBLEM

Cat owners have no easy way to detect early signs of health issues from litter box behavior.

Cats often hide symptoms of illness such as urinary blockages or weight loss. Subtle changes in litter box visit frequency, duration, and body weight are hard to notice without consistent tracking. By the time owners spot a problem, the condition may already be serious.

BY THE NUMBERS

64.5%

of cats with urinary tract issues show no obvious symptoms until the condition becomes serious

INTENDED USER GROUP

Cat owners who want passive, automatic monitoring of their cat's health through litter box usage patterns, without relying on manual observation or frequent vet visits.

Our Solution & Key Features

IN ONE SENTENCE

A litter box mounted edge device that uses a PIR sensor and load cell to track each visit's duration and weight, runs anomaly detection against a personalized baseline on the ESP32S3, and syncs logs to a cloud dashboard for long term trend viewing.

HARDWARE STACK

XIAO ESP32S3 | Load cell + HX711 | PIR sensor

KEY FEATURES

01 Visit Detection

PIR sensor captures entry and exit automatically for every visit.

02 Weight Tracking

Load cell measures cat weight per visit and tracks long term trends.

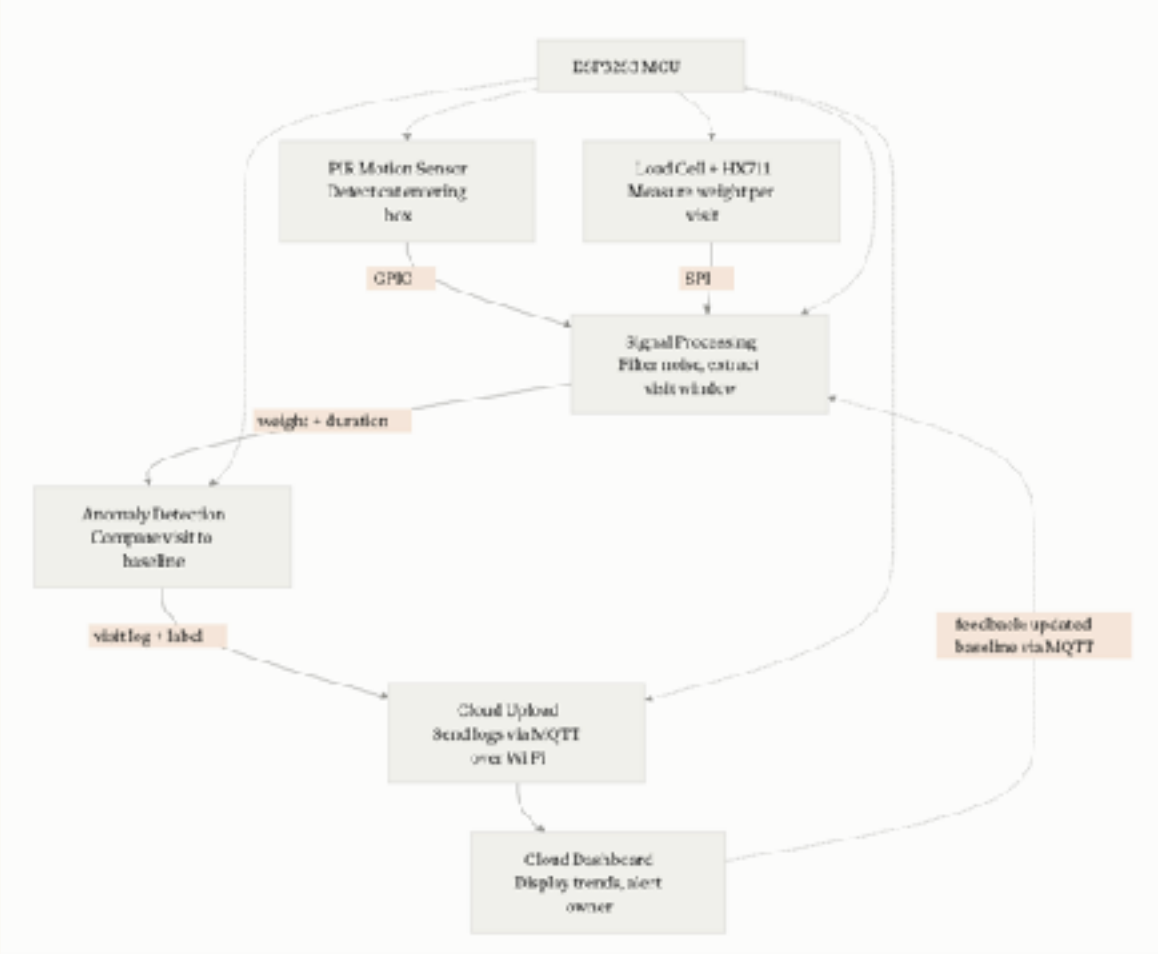
03 On Device Anomaly

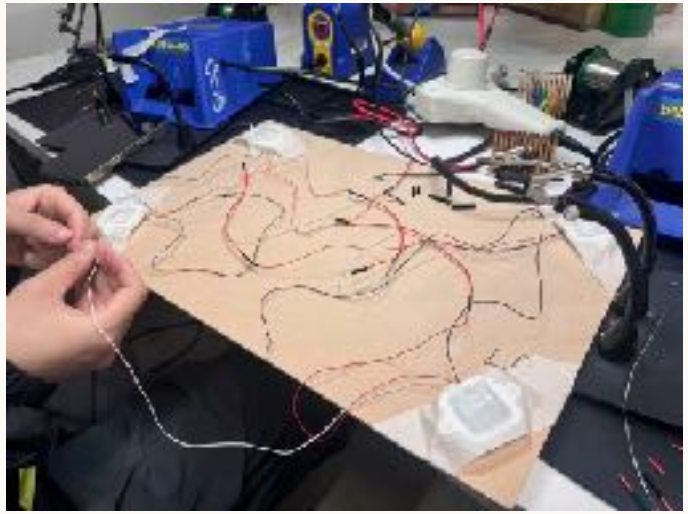
Rule based detector compares each visit to a personalized 7 day baseline.

04 Cloud Trends

MQTT upload to a dashboard so owners can view history and alerts.

Updated System Architecture





Solid lines mark tested interfaces. Dashed lines mark interfaces not yet verified. Update block status as components move from in progress to proven.

Component Validation & Results

Each component validated with a quantitative test. Evidence: photos, plots, and commits.

01

YUNA XIONG

Sensor Hardware

METHOD

Connected the HX711 load cell to the XIAO ESP32S3 and placed a 201.56 g reference weight 20 times. Each reading was recorded after a 5000 ms stabilization window at 1000 ms intervals.

RESULT

Mean error 0.11 g, standard deviation 0.23 g. GO verdict, well within the 10 g accuracy threshold.

02

MENGQI SHI

Signal Stability

METHOD

Analyzed the 20 trial weight readings from the HX711 sensor to evaluate signal noise and reading consistency. Computed error distribution, peak to peak range, and standard deviation across all trials.

RESULT

Peak to peak error range 0.72 g, max error 0.43 g, min error 0.29 g. Signal stable enough to serve as input for the anomaly detection pipeline.

03

XIN LUO

Data Logging and Transmission

METHOD

Developed the ESP32S3 firmware (Read_ESP32S3.ino) to read and log weight data via serial output. Verified that all 20 readings were captured without data loss at 1000 ms intervals over the test session.

RESULT

20 out of 20 readings successfully captured with zero dropped samples. Firmware logging validated for future MQTT cloud upload integration.

Budget Update

TOTAL BUDGET

$ 350

SPENT SO FAR

$ 57.38

REMAINING

$ 292.62

PURCHASES TO DATE

Item	Qty	Unit Cost	Total
[LiPo battery]	[1]	[$ 5.83]	[$ 5.83]
[LiPo charger module (TP4057, 5pcs)]	[1]	[$ 7.56]	[$ 7.56]
[Cat litter box(Smart Paws,enclosed)]	[1]	[$ 43.99]	[$ 43.99]
[XIAO ESP32S3]	[2]	[$ 0.00]	[$ 0.00]
[Load cell + HX711]	[1]	[$ 0.00]	[$ 0.00]
[PIR motion sensor]	[1]	[$ 0.00]	[$ 0.00]